Teaching Kernel SVM: linear, polynomial and RBF kernel comparison	
Student Name: Neeraj Yelahanka Mallikarjuna	**Student Id -** 24160010

1. Introduction

Support Vector Machine: SVM is a supervised machine learning procedure for classification, regression, and outlier detection. In classification, the basic concept is that you want to find such a decision boundary that the classes are separated, but the margin between the decision boundary and the nearest training patterns is as large as possible. People nearest to those points are call support vectors and set up the direction of the boundary Cortes SVM to be important as SVM is an algorithm that often technically performs well with small and medium sized structured data sets, especially if the number of features is moderate and high values and if there is a good class separation.

The primacy teaching focus of this tutorial is that of kernel. A kernel provides SVM with an opportunity to deal with data which are not linearly separable by implicitly mapping the original features into a higher dimension in which case a separating line may be possible In fact, this is equivalent to stating that changes in the choice of kernel change the flexibility of the classifier. This tutorial covers the differences between the linear, polynomial and the radial basis function, or RBF, kernels - explains them with code and shows what the hyperparameters do with regards.

2. Data set and experimental design

The actual dataset used in this tutorial is the UCI Machine Learning Repository's Seeds dataset. It has 210 observations, 7 numerical features, and three wheat varieties, with no missing value published on the repository page (UCI Machine Learning Repository, n.d.). This makes it appropriate for a shorter tutorial because there is a big enough difference between the models, but also small enough for the entire workflow to be quickly grasped.

The notebook consists of initial steps of loading the dataset, checking the balance of classes, and checking relationships between features. The target variable, wheat variety has input features some geometrical values or as features for example area, perimeter, compactness, and groove length of kernels. The data are then divided into a training set and test set. This step is important because the model needs to be evaluated on data that it has not been fitted to yet, rather than on the data used to fit the model.

3. How the three kernels differ

The simplest kernel to use is the linear kernel. It tries to separate the classes using a perpendicular line in the original feature space: As we know "when data is already very close to being linearly separable, the linear kernel tends to work well while it eases to interpret and from the computational perspective it is cheap to operate than other more flexible kernels" it's really the best place to go to because it brings the main SVM concept to the foreground without any further complications.

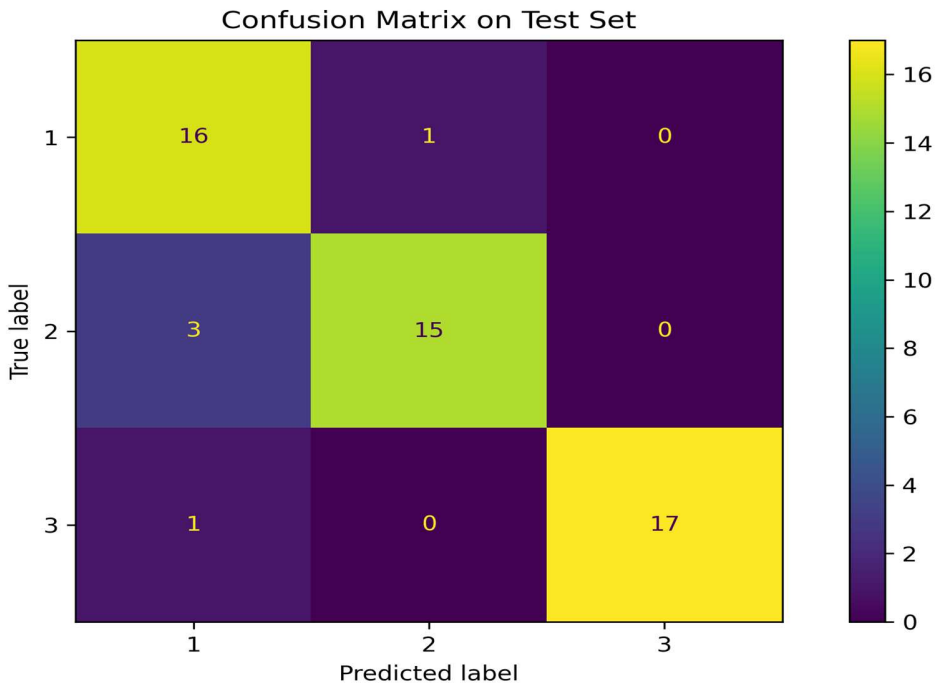
The polynomial kernel means that the decision boundaries are curved. Instead of updating just the original features, it will model interactions to some degree. This can be helpful when the boundaries for the classes are not a straight line but do have a structured shape.

The most flexible out of all three is the RBF kernel. non-linear regions of the decision space, and is often a good default, if there is no reason to believe linear separation is possible. In the scikit-learn implementation of SVC the RBF is used many times because of the fact that it has an ability to learn the shape of the boundary between classes while highlighting the points without proper knowledge about the interactions of features and as with explicit approach in which interaction of features are defined by hand scikit-learn.

4. Confusion Matrix.

The model performance, a confusion matrix, was performed for the best tuned SVM model with the test set. The confusion matrix compares the actual class label and the predicted class label and gives an indication of how well the model separates the three varieties of wheat across the Seeds dataset.

Most observations were classified correctly, and errors were extended over the few observations which support the overall test accuracy of 0.9057.This is important as accuracy alone cannot indicate where mistakes are being made, but the confusion matrix can give a better view of model performance.



5. Results from actual data set

The best tuned model, in this notebook a linear SVM is used with C = 100. The average cross validation accuracy results were found to be 0.9681 and the test (held out) accuracy results were found to be 0.9057. This result has a nice teaching point; it suggests that a more complex kernel is not necessarily a better kernel. Even though SVM comes close to being the best model for this dataset, this run was a linear boundary along with stronger control.

The following example message is directly from the kernel and supports the same message. On the linear and RBF configurations of the standard worked about the same while the polynomial was a bit worse. This gives us a hint that in the Seeds dataset we have a class structure that we can already separate well in the original feature space, as it is scaled.

The confusion matrix is also useful in a teaching tutorial as the confusion matrix displays where the errors are occurring, rather than just one overall score. When classes are close together, a confusion matrix is useful to understand which classes are hardest to tell apart and if misclassifications will be symmetric or concentrated in one class.

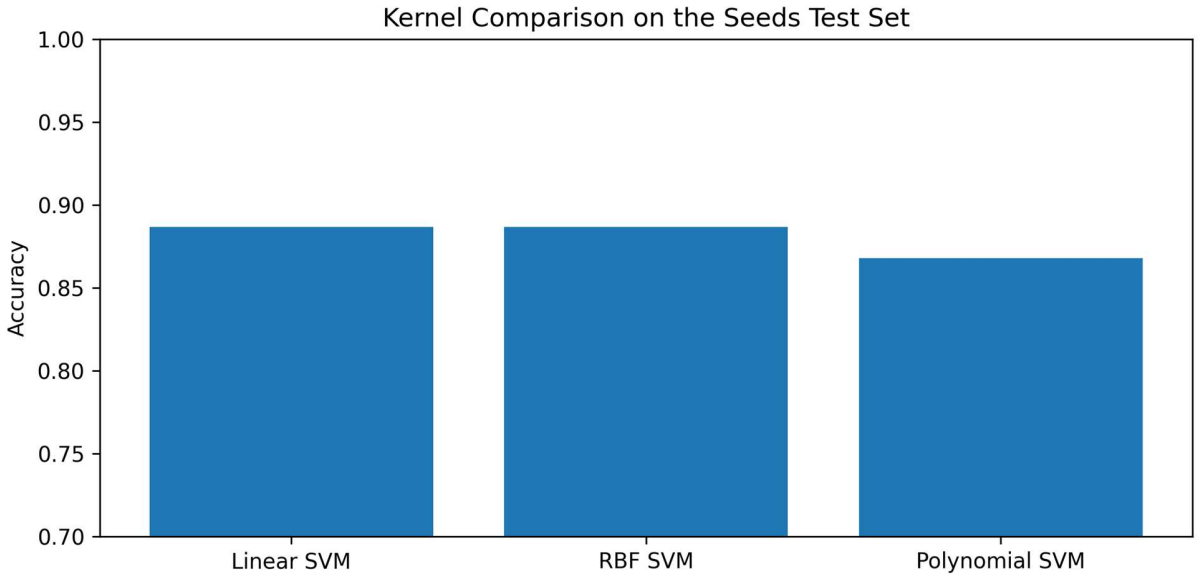


Figure 1. Assess how accurate you are with three types of kernels on the Seeds dataset.

6. Why does the synthetic moons dataset include

The Seeds dataset is realistic, but it is not very visual in demonstrating kernel behavior since it is 7-dimensional. For this reason, the synthetic dataset of moons is also used in this tutorial. This datum set is not the main experiment, rather, a teaching tool. It has a curved structure, so we have the easy instrument to see the different decision boundaries between different kernels.

On the moons plot the linear kernel leads to a straight line as border follow the curved shape of the classes. The polynomial kernel is one way of capturing some of that curvature, but the RBF kernel is a much more natural non-linear line. This visual comparison is good to go from a purely abstract concept of the kernel trick to pictures.

This visual demonstration makes the tutorial stronger, as well as making the choice of kernels easier to understand for beginners. It also supports the comparison between model interpretability and model flexibility, which is an important part of the process for applying SVM effectively. Therefore, the data set on the moons adds value to the teaching aspect of the tutorial even though the main evaluation is from the Seeds data set.

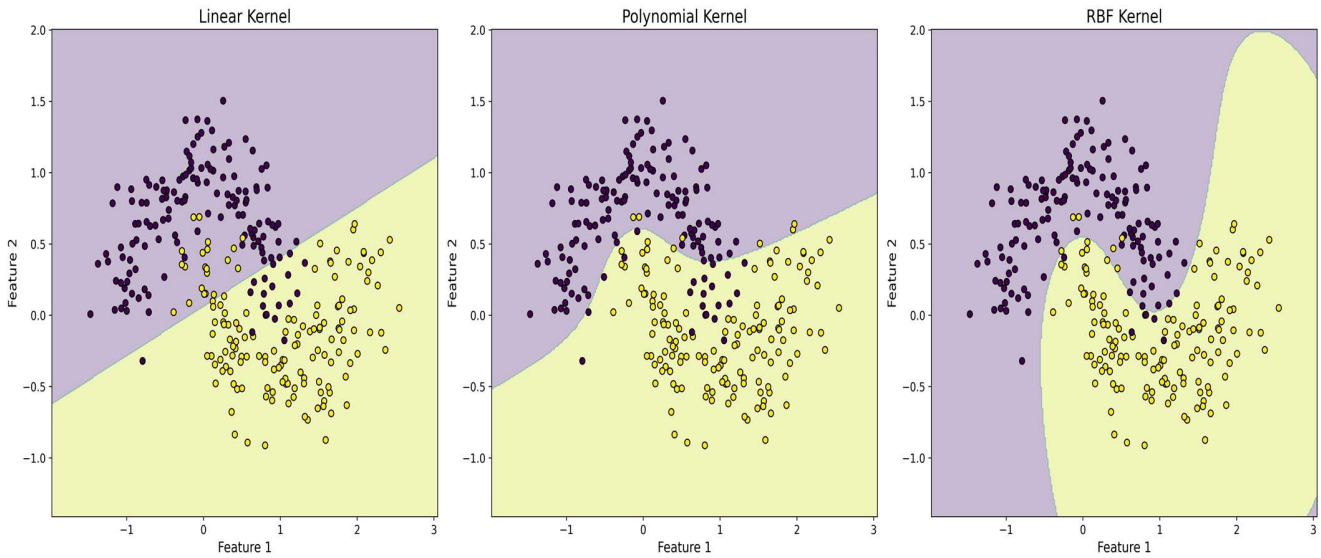


Figure 2. Decision boundaries artificially created using linear, polynomial, and RBFs kernels dataset of moons.

7. The effect of C and gamma

Two hyperparameters are especially relevant to the use of SVM; that is, C and gamma.

C can control the balance between maximizing the margin and penalizing the classification of errors. Two smaller C means a wider margin and can find more errors by training data may help to generalize the model but on the other hand it can be underfitting. A big C has a bigger penalty for errors, and this tends to lead to a tighter boundary which is a better fit to the training.

Gamma is only relevant for non-linear kernels. It controls the degree to which the impact of its individual training example is perceived. A low gamma is used to get smoother and wider decision regions. A high gamma produces very local highly flexible boundaries which can overfit the noise. In the plots of the tutorial, for low gamma, the boundary is much smoother, whereas, for high gamma, the shape is much more irregular.

These plots are especially important since they allow us to teach a practical lesson: Model performance is not only determined by kernel choice. It is very possible to have a good kernel with bad hyperparameters, or one with simple kernel and the right hyperparameters and perform much better.

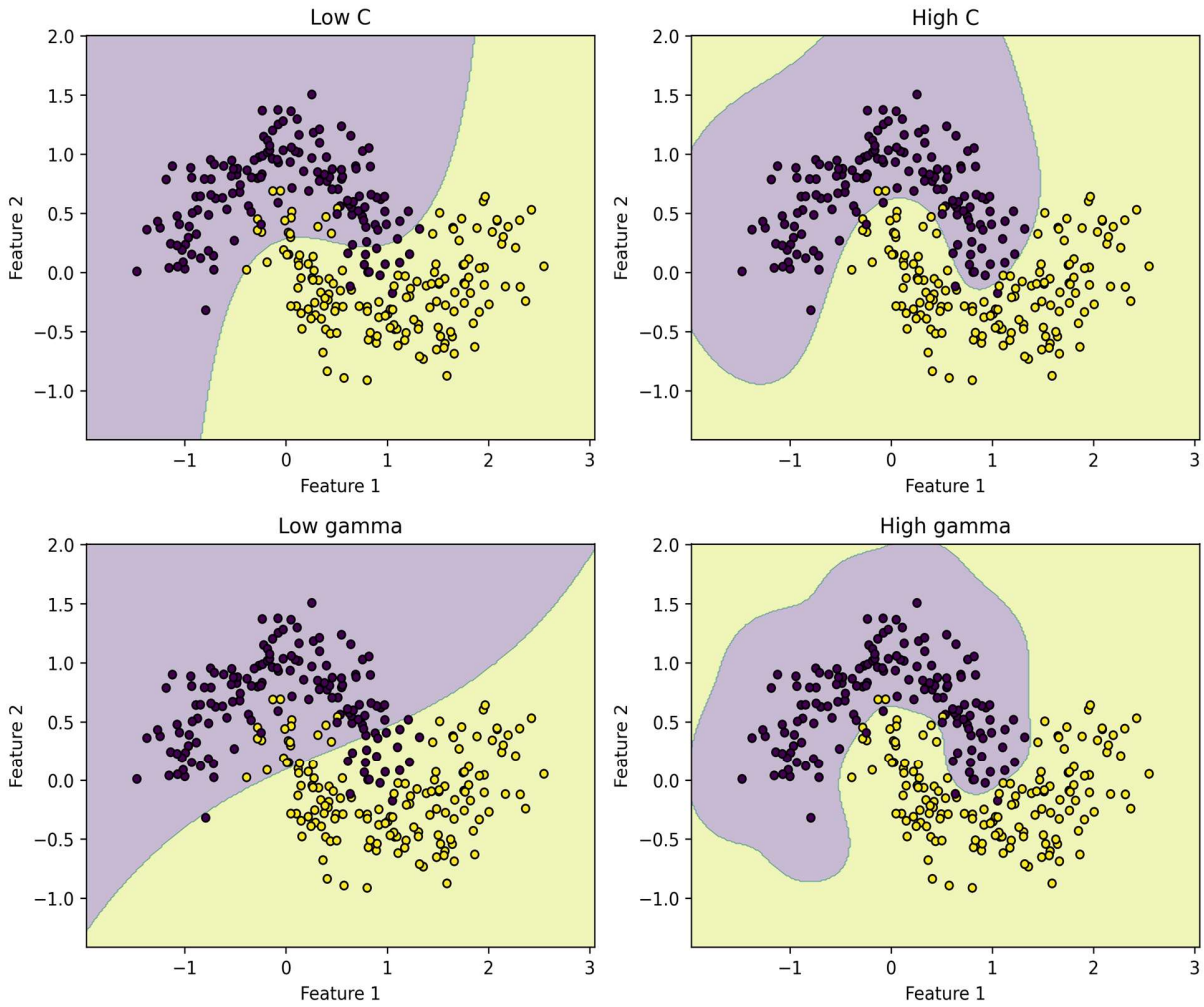


Figure 3. Illustration of the effect of C and gamma on flexibility of RBF decision.

8. Feasible hints when using SVM in a kernel

If someone wants to use SVM in their own work, a reasonable starting process is:

1. Standardize the features.
2. Begin with a linear kernel as a baseline.
3. If the linear boundary seems too simple, test RBF and polynomial kernels.
4. Use cross-validation, not only one test split, to compare models.
5. Tune C and, where relevant, gamma and degree.

6. Evaluate with more than one metric, including a confusion matrix or class-wise scores.

This approach keeps the method teachable and reproducible. It also matches professional machine learning practice, where baseline models are compared against more complex alternatives rather than assuming the most sophisticated option must be best.

9. Ethical considerations- AI & accessibility

Although this tutorial is only dealing with accuracy and decision boundaries, this metric of machine learning is also one which should be judged from an ethical perspective. A small data set may not represent larger populations and actual deployment conditions, and high-test score may not represent fairness, robustness, or social usefulness. In actual applications, the results of models should be able to inform human judgement but not vice versa, but especially in high-stakes areas.

Accessibility is an issue regarding tutorial design. Plots must have good labels and understandable font sizes and explanations that do not just use color. Clearly designed heading structure of the notebook and the pdf should enable the audience to follow the flow of work in a step-by-step way. This is particularly important as the reason for the assignment is not necessarily to run code for the sake of running code, but to teach the method effectively.

10. Conclusion

This tutorial sets out to teach the change in behaviour of Support Vector Machines when the choice of the kernel is changed. The important thing to learn is that kernels are the things that control the flexibility of a model. The linear kernel is simple and can be surprisingly strong. The polynomial kernel introduces structured curvature into the situation, but the RBF kernel provides the most (but also the greatest) flexibility in deciding whether some pattern is non-linear.

In the notebook used for this tutorial, the best model that was tuned with the Seeds dataset was a linear SVM model with C = 100 and was achieved via a cross-validation accuracy of 0.9681 and a test accuracy of 0.9057. This result underlines an important tenet in machine learning such as not necessarily assuming complexity to add to a machine learning algorithm but learning from the data.

By using a real data set and construction of synthetic teaching data set in tandem, both practical performance and visual intuition are obtained in this tutorial. For learners, such kernel SVM introduces a combination that may make it easier to understand, at the same time also easy to apply in future work.

11. References

Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273-297.

https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html

scikit-learn. (n.d.-b). Pipeline.

https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html

scikit-learn. (n.d.-c). Support vector machines.

https://scikit-learn.org/stable/modules/svm.html

UCI Machine Learning Repository. (n.d.). Data set in Seeds

https://archive.ics.uci.edu/ml/datasets/seeds